3.3. Alternative facilities for protecting shared data

3.3.1. Protecting shared data during initialization

3.3.2. Protecting rarely updated data structures

3.3.3. Recursive locking

3.4. Summary

Chapter 4. Synchronizing concurrent operations

4.1. Waiting for an event or other condition

4.1.1. Waiting for a condition with condition variables

4.1.2. Building a thread-safe queue with condition variables

4.2. Waiting for one-off events with futures

4.2.1. Returning values from background tasks

4.2.2. Associating a task with a future

4.2.3. Making (std::)promises

4.2.4. Saving an exception for the future

4.2.5. Waiting from multiple threads

4.3. Waiting with a time limit

4.3.1. Clocks

4.3.2. Durations

4.3.3. Time points

4.3.4. Functions that accept timeouts

4.4. Using synchronization of operations to simplify code

4.4.1. Functional programming with futures

4.4.2. Synchronizing operations with message passing

4.5. Summary